

CHANDRA SEKHAR BHAMIDI

Applications Engineer | 7nm CPU Implementation | Timing Analysis

Tokyo, Japan • Valid Work Visa (Jul 2030) • No Sponsorship Required

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PROFESSIONAL SUMMARY

Applications Engineer at Synopsys Tokyo with hands-on experience in advanced-node physical design, timing analysis, and implementation methodologies. Worked on a 7nm ARM-based CPU core (3.5M+ instances, 350+ macros), contributing to floorplanning, timing optimization, routing, and setup closure using Fusion Compiler, StarRC and PrimeTime. Currently based in Tokyo with valid work authorization through July 2030.

PROFESSIONAL EXPERIENCE

Applications Engineer — Synopsys Tokyo

Nov 2024 – Present | Tokyo, Japan

Tools: Fusion Compiler · IC Compiler II · PrimeTime | Technology: 7nm FinFET | Project: ARM-based CPU Core

- Contributed to physical design and setup timing closure for a 7nm ARM CPU core (3.5M+ instances, 350 macros, 9 metal layers) using Fusion Compiler, PrimeTime, and StarRC.
- Contributed to achieving 2.54 GHz (pre-DFT) and 2.47 GHz (post-DFT) against a 2.6 GHz target through setup timing optimization across worst setup corners.
- Explored 20+ floorplan variants to improve macro locality, reduce critical-path wirelength, and enhance routability for a 7nm CPU implementation.
- Analyzed timing sensitivity across worst setup corners through clock uncertainty sweeps at compile, CTS, and route stages, analyzing WNS/TNS, runtime, and timing margin tradeoffs.
- Improved timing/DRC convergence using incremental placement and route_auto tuning, identifying QoR degradation beyond optimal routing effort.
- Improved setup closure by prioritizing optimization on the two most timing-critical path groups to accelerate WNS convergence during late-stage timing optimization.
- Isolated DFT-specific timing interactions during functional-mode closure by constraining newly introduced scan/test ports, reducing non-functional timing noise and improving setup convergence prior to scan-mode analysis.
- Performed timing analysis using PrimeTime and executed manual timing ECOs — cell upsizing, VT swaps, and buffer insertion — targeting setup violations on top-10 critical paths post-routing.
- Developed TCL script to filter and rank worst-violating path groups by WNS, reducing timing-debug turnaround time by 30%.

Digital Implementation Trainee — ZOICX Technologies

Jun 2024 – Oct 2024 | Hyderabad, India

Tools: Design Compiler · IC Compiler II · Formality | Project: ARM Core | Technology: 28nm/9 layers

- Executed full physical design flow on ARM Core at 28nm — 300,000+ gates, 29 macros, 1 clock domain — achieving 500 MHz target.
- Performed logic synthesis using Design Compiler; verified 300,000-gate post-synthesis netlist equivalence using Synopsys Formality shell. Resolved missing via violations during floorplanning; ran 10+ trials with varying core shapes and utilization targets to eliminate routing congestion.
- Benchmarked 3+ open-source EDA tools (OpenROAD, Yosys, OpenSTA) against Synopsys commercial flow across synthesis, P&R, and STA stages.

Physical Design Engineer – Trainee — RiseTime Semiconductor

Oct 2023 – Mar 2024 | Hyderabad, India

Tools: Design Compiler · IC Compiler II · Formality | Project: ARM Core | Technology: 28nm/9 layers

- Executed synthesis-to-post-route physical design flow for a 72K-gate design with 6 macros and 2 clocks at 454 MHz, covering synthesis, floorplanning, placement, CTS, routing, and timing checks.
- Resolved missing-via, pin-density, and congestion issues through keepout-margin optimization and multiple floorplan trials with different core shapes and utilization targets.

TECHNICAL SKILLS

Physical Design	Floorplanning, macro placement, placement optimization, routing optimization, congestion reduction, DRC closure, setup timing convergence
Timing / STA	PrimeTime, WNS/TNS debug, path-based analysis (PBA), ECO timing closure, VT swap, buffering, clock uncertainty sweep
EDA Tools	Fusion Compiler, PrimeTime, IC Compiler II, Design Compiler, StarRC, Formality
Methodology	QoR optimization, incremental placement, uncertainty sweep, timing sensitivity study, DFT timing constraints
Automation	TCL scripting, timing report parsing, QoR tracking, path filtering
Technology	7nm / 28nm (TSMC) 3 PVT corners 9–13 metal layers

EDUCATION

Post Graduate Diploma — Physical Design

2022 – 2023

Sumedha IT, Hyderabad | ALU Design (RTL-GDSII with Synopsys Tools)

B. Tech — Electronics & Communication Engineering

2018 – 2022

JNTU, Hyderabad | Publication: IEEE — J-Pole Antenna for ADS-B Signals (Apr 2022) | NPTEL Certified: VLSI Design Flow RTL to GDS